DOMINO

Family Resilience Intelligence System

“When one thing changes, everything connects.”



Concept rendering: entities of the fictional Amina household

6th Annual COE STEM Programming with AI Competition · AI for Family

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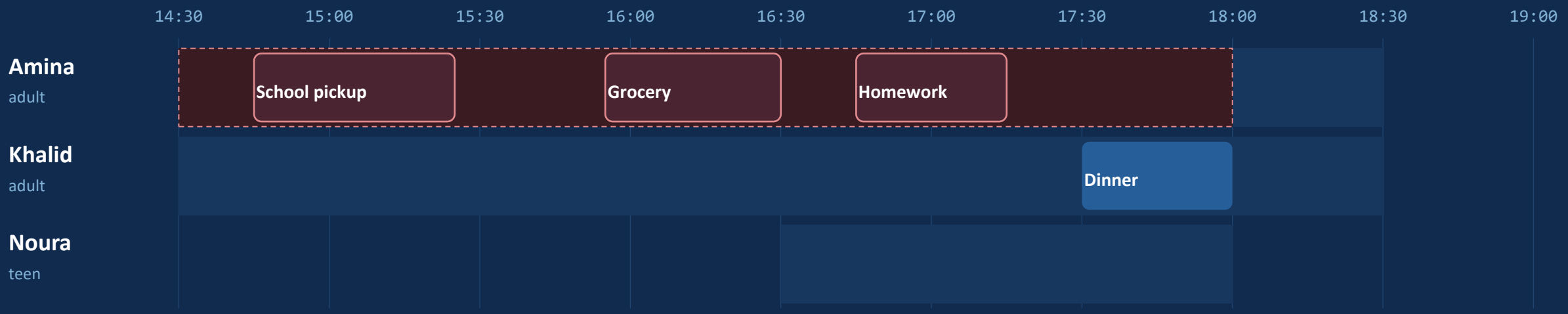
ADNOC Schools — Madinat Zayed

One disruption, multiple consequences

At 2:30 PM, Amina becomes unavailable until 6:00 PM.

DISRUPTION · 14:30

Original plan (fixture) · shaded = available · hatched = unavailable



School pickup · Tier 0 · essential

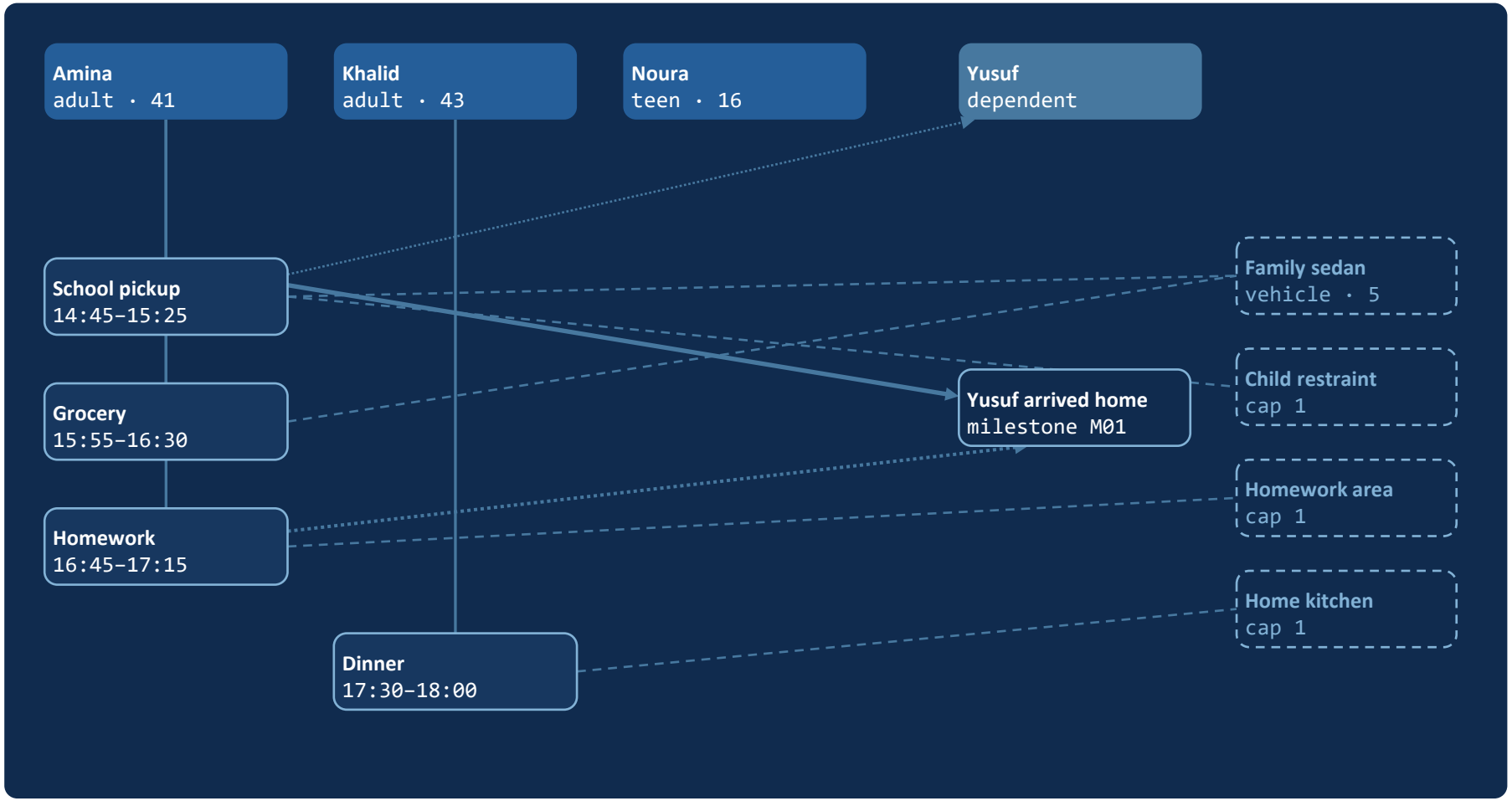
Homework assistance · Tier 1

Grocery collection · Tier 3

Dinner (Khalid) is not obviously affected.

Source: data/households/amina.json (fictional household). Times are Asia/Dubai, simulated date 29 Oct 2026.

The hidden system behind family life



solid = assigned to / enables dashed = uses resource dotted arrow = depends on / serves dependent

Source: Amina fixture; edge types from backend/app/services/dependency_graph.py. Native, editable diagram.

3 responsibilities → 1 caregiver

Pickup, groceries and homework are all assigned to Amina.

A shared vehicle

Pickup and groceries both need the same sedan, one at a time.

A gating milestone

Homework can only happen after “Yusuf arrived home”.

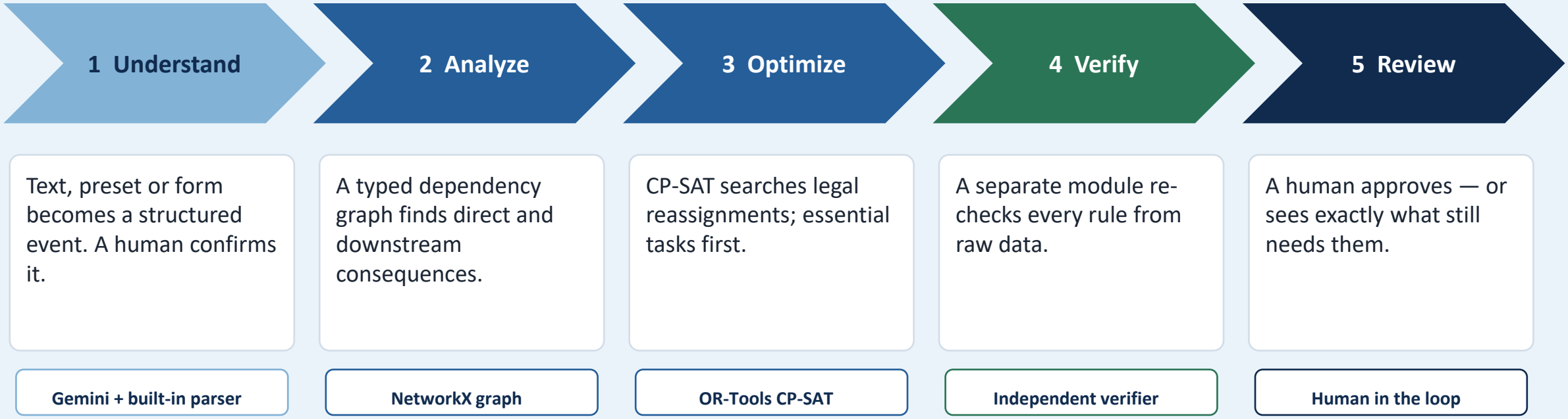
Why calendars and task lists are not enough

Function	Typical calendar / task list	DOMINO	Slide
Records scheduled events	**Yes — its core purpose**	✓ **Household model with baseline schedule**	S07
Models dependency relationships	Typically limited or absent	✓ **Typed graph: assigned-to, enables, depends-on …**	S12
Detects cascading consequences	Typically limited or absent	✓ **Direct and downstream impact with causal paths**	S17
Calculates feasible alternatives	Typically limited or absent	✓ **CP-SAT optimization with priority tiers**	S21
Verifies constraints	Typically limited or absent	✓ **Independent 18-rule verifier**	S23
Supports counterfactual comparison	Typically limited or absent	✓ **Recomputed what-if side by side**	S19

This compares common scheduling functions with the broader requirements of recovering from a disruption. Individual products vary; no claim is made about any specific product.

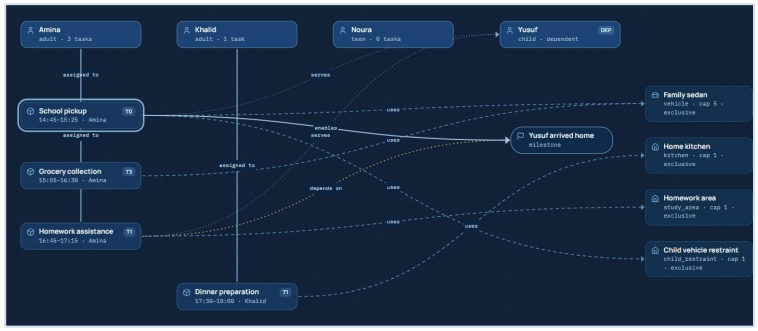
The DOMINO solution

DOMINO calculates how a family can recover when its original plan becomes impossible.



DOMINO simulates the consequences of specified, hypothetical events. It does not predict the probability of emergencies.

Three households, different constraints

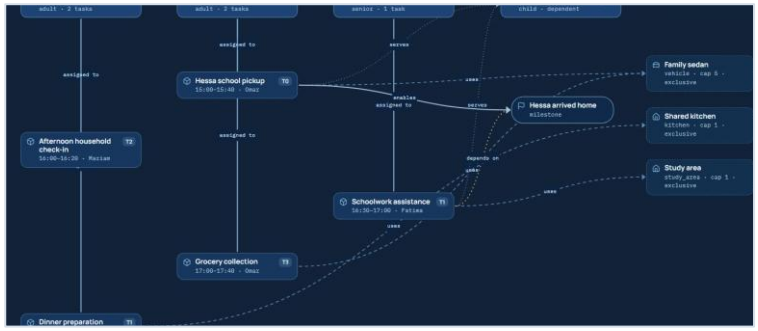


The Amina Family

4 members · 4 tasks · 4 resources

Two authorized adult drivers; a teen who may help with homework but cannot drive; a 9-year-old dependent.

Exercises: clean reassignment

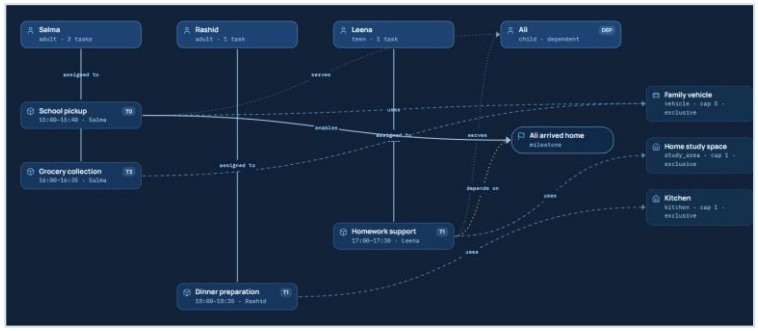


The Al-Mansoori Multigenerational Family

4 members · 5 tasks · 3 resources

Three generations; a 70-year-old grandmother helps with schoolwork and check-ins but holds no driving permission.

Exercises: several valid alternatives



The Salma Family

4 members · 4 tasks · 3 resources

One licensed driver, one vehicle: losing either leaves school pickup without a legal replacement.

Exercises: honest infeasibility

Source: data/households/{amina, multigenerational, restricted-transport}.json — fictional; screenshots from the published prototype, 28 Oct 2026.

The Amina family baseline

A

Amina · 41 · adult

drive · pickup_authorized · homework_assist · shopping

K

Khalid · 43 · adult

drive · pickup_authorized · shopping · meal_prep

N

Noura · 16 · teen

homework_assist only · restrictions: drive, pickup_authorized

Y

Yusuf · 9 · dependent

never assigned tasks · authorized carers: Amina, Khalid, Noura

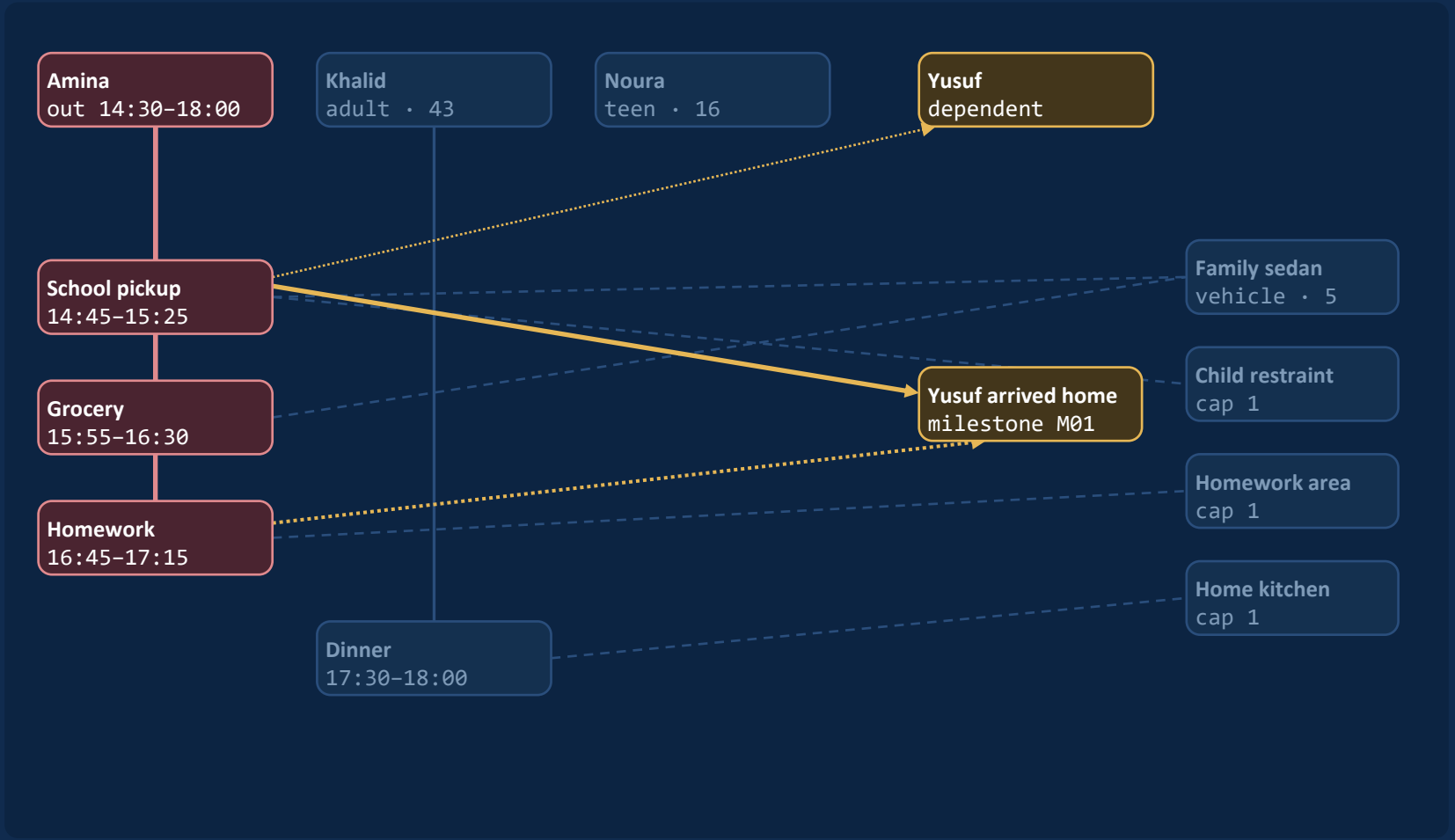
Responsibility	Tier	Baseline	Time	Needs
School pickup	0	Amina	14:45–15:25	drive, pickup_authorized, sedan, child restraint
Grocery collection	3	Amina	15:55–16:30	shopping, sedan
Homework assistance	1	Amina	16:45–17:15	homework_assist, study area, milestone M01
Dinner preparation	1	Khalid	17:30–18:00	meal_prep, kitchen

```
T01 pickup ──enables──▶ M01 "Yusuf arrived home"
T03 homework ──depends on──▶ M01
```

The dependency is stored in the household model, not inferred from a picture. Baseline check: 18 verifier rules, 0 violations (test T01).

Source: data/households/amina.json; benchmark case T01 "Amina baseline".

The cascade begins



Entity	Before	After
Amina	available	unavailable
School pickup	valid	direct
Grocery	valid	direct
Homework	valid	direct + risk
Yusuf arrived home	satisfiable	at risk
Yusuf	served	at risk
Dinner, Khalid, Noura	valid	unaffected

Real causal path (engine output):
Disruption → Amina → School pickup → Yusuf arrived home → Homework
[disrupted_by › assigned_to › enables › depends_on]

4 direct · 2 indirect · depth 3 · weighted coverage 100% → 25%

Source: POST /v1/simulations on the live API, 28 Oct 2026; direct = Amina + 3 tasks, indirect = milestone + Yusuf.

From confirmed event to reviewed recovery

LANGUAGE	GRAPH	GRAPH	CP-SAT	VALIDATOR	HUMAN
1 **Confirmed event**	2 **Dependency analysis**	3 **Candidate generation**	4 **Constraint optimization**	5 **Independent validation**	6 **Recovery review**
1 event · Amina · 14:30–18:00 · confirmed by user	4 direct · 2 indirect · depth 3	56 candidates after static filtering	OPTIMAL · 5 lexicographic passes	56 checks · 18 rules · 0 violations	approval needed · pickup consent pending

The language model never sees the solver. The solver never marks its own homework.

Language AI interprets text only. Everything that decides or checks an assignment is deterministic, tested code.

Source: live flagship run (POST /v1/recovery-plans), 28 Oct 2026. Counts are measured outputs, not targets.

What makes DOMINO distinctive

1 Structured family digital twin

Members, skills, explicit permissions, time windows, resources and milestones as validated data.

2 Graph-based cascade analysis

Typed edges show what a change touches directly and downstream, with causal paths.

Household digital twin

3 Mathematical recovery optimization

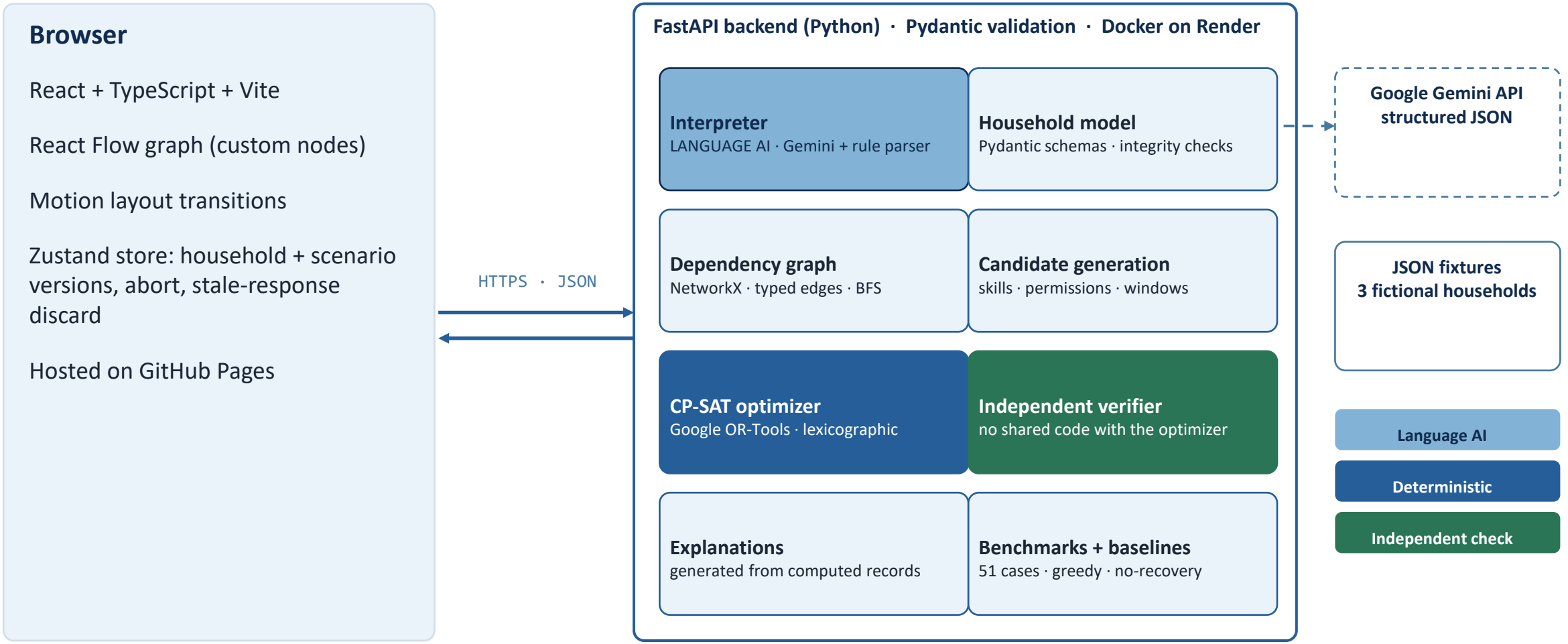
CP-SAT searches legal arrangements; essential responsibilities are protected first.

4 Independently verified decisions

A separate validator re-checks every rule; unresolved tasks stay visible.

The value is in the combination: a chatbot cannot guarantee a permission; a solver without a model does not know what to re-plan.

System architecture



Source: repository structure (backend/app/services/, frontend/src/*); deployment per render.yaml and pages.yml. Native, editable diagram.*

The household digital twin

FamilyMember

id · name · age · role · isDependent
skills[] · permissions[] · restrictions[]
availability[{startMin,endMin}]
authorizedDependentIds[]

HouseholdTask

id · tier 0–3 · priorityWeight · durationMin
earliestStart · latestEnd · fixedStart?
requiredSkills[] · requiredPermissions[]
requiredResourceTypes[] · requiredConsent
predecessorTaskIds[] · requiredMilestoneIds[]

HouseholdResource

id · name · type · capacity · exclusive
availability[] · unavailableWindows[]
allowedMemberIds? (vehicle, kitchen, …)

Assignment

taskId → memberId
startMin · endMin · resourceIds[]
changedFromBaseline

Milestone

id · title
enabledByTaskIds[]
dependentId? e.g. “Yusuf arrived home”

Disruption (5 types)

member_unavailable · resource_unavailable
task_rescheduled · availability_changed
resource_capacity_changed
affectedEntityIds[] · window · confirmed

P01 Amina · adult · skills [driving, shopping, homework_assist] · permissions [drive, pickup_authorized] · authorized for P04 · available 14:30–18:30

Source: backend/app/models/domain.py; example from data/households/amina.json. Simplified diagram: fields abbreviated.

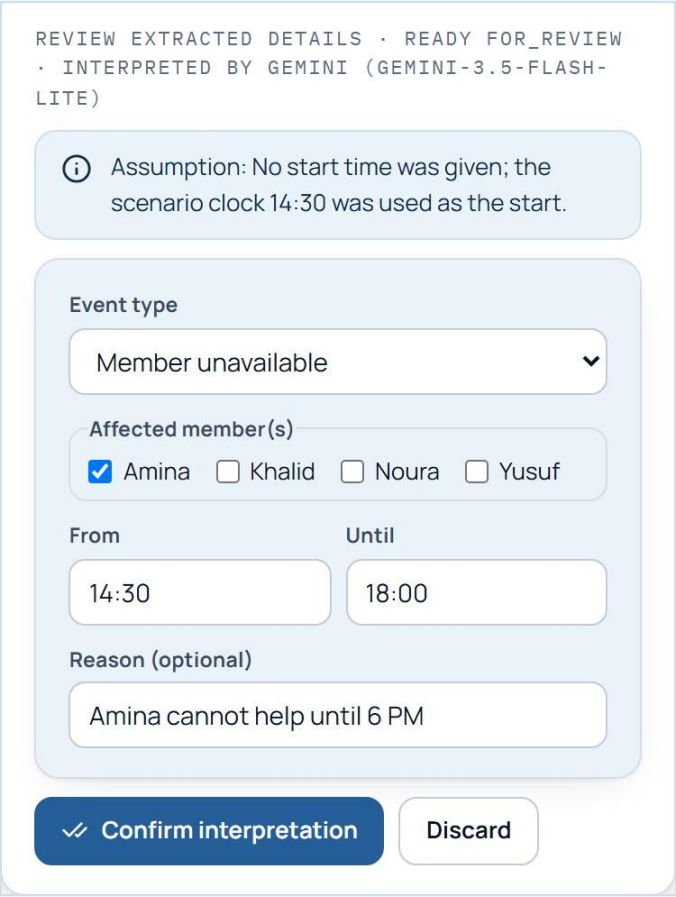
Natural-language interpretation

1. **Natural-language description**
2. **Structured interpretation**
3. **Entity resolution**
4. **Time extraction**
5. **Schema validation**
6. **Human confirmation**

Gemini: step 2 only

```
Observed output (live API)

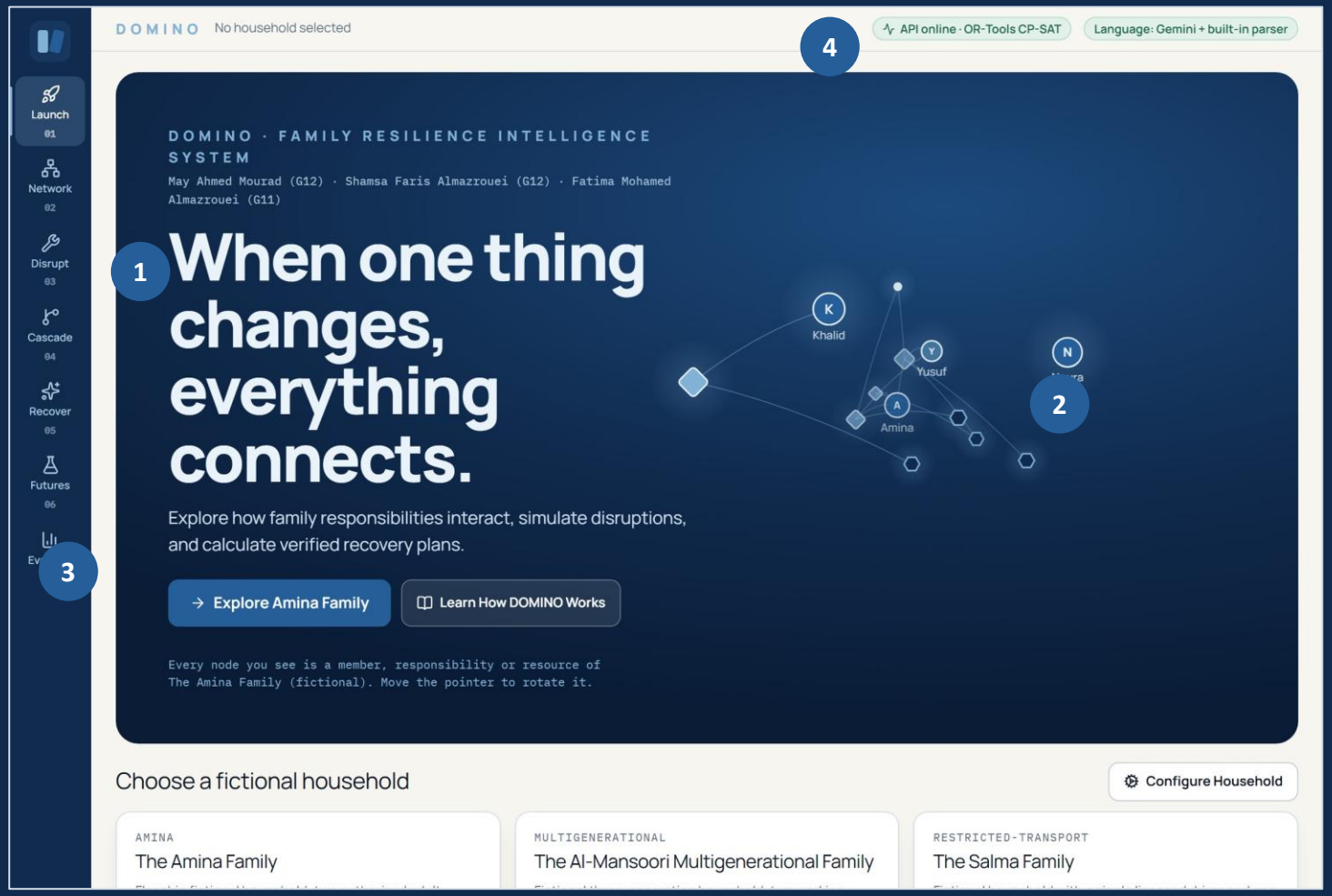
“Amina cannot help
 until 6 PM”
   | clock 14:30
   ▼
{
  "type": "member_unavailable",
  "affectedEntityIds": ["P01"],
  "startMin": 870,   // 14:30
  "endMin": 1080,   // 18:00
  "confirmed": false
}
engine: gemini-3.5-flash-lite
assumption: start = scenario
clock 14:30
```



Gemini interprets. It never authorizes or schedules.

Source: backend/app/services/interpreter.py + rule_parser.py; live probe 28 Oct 2026; accuracy in TESTING.md (12-sentence sets).

System launch

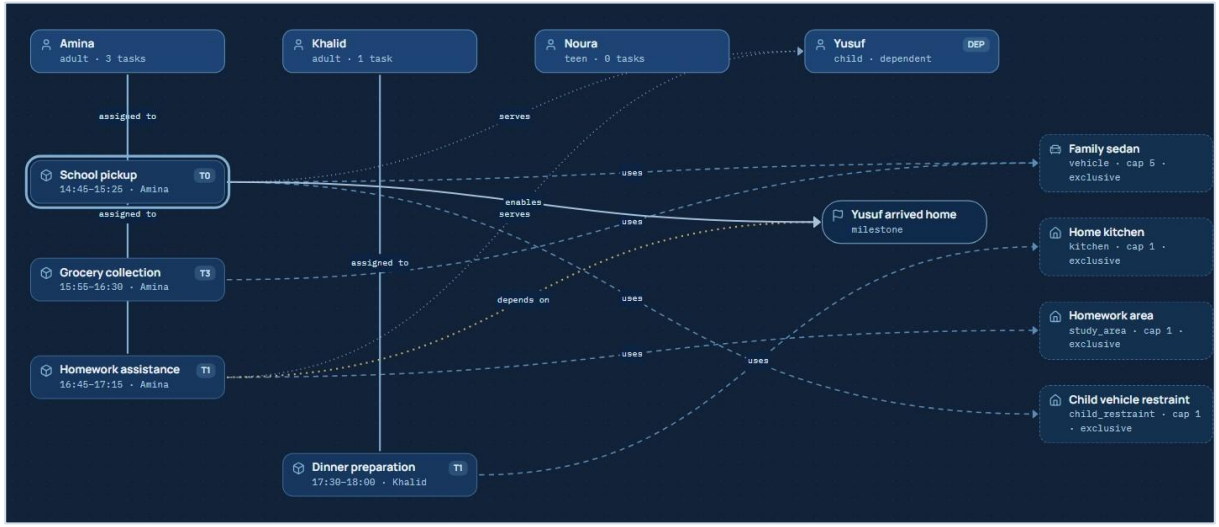


Real screenshot of https://maymourad.github.io/DOMINO/ captured 28 Oct 2026. Status chips show the state at capture time only.

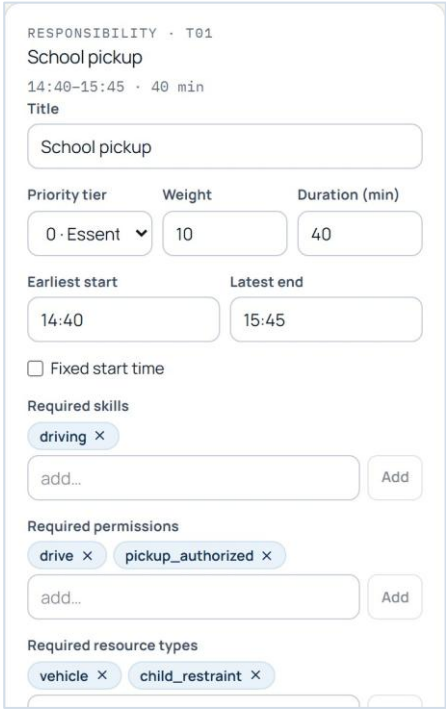
1. Central proposition; project identity and the team credited under the title
2. Interactive constellation: real entities of the selected household
3. Navigation across the seven screens
4. Live status: API online · OR-Tools CP-SAT · language engines

Below the hero: three fictional households and Configure Household (editing with live validation).

Family Network



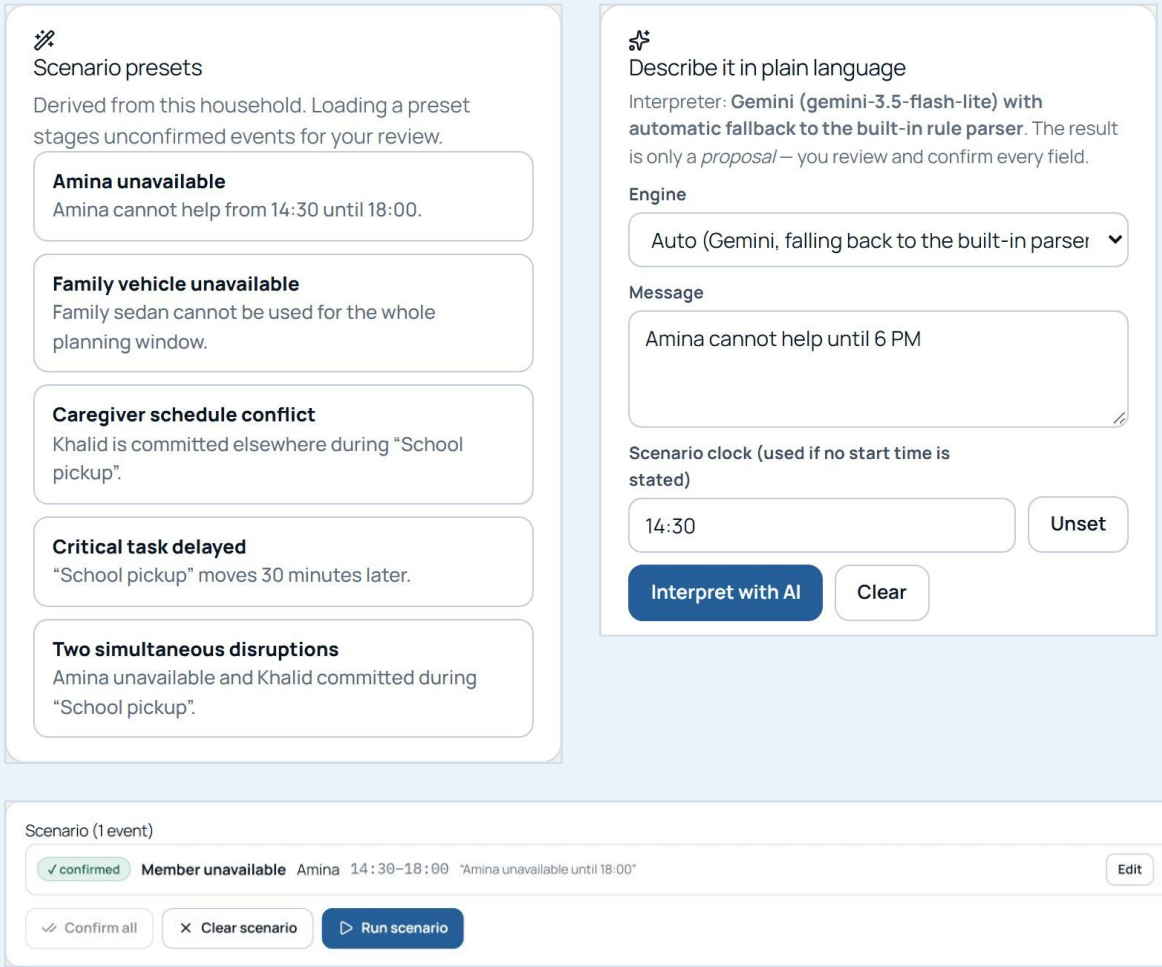
Toolbar: add task · add member · add resource · remove · undo · reset to the validated baseline. Select School pickup (T01) to edit it; the household is re-validated by the backend after every edit.



1. Custom typed nodes: people, tasks, resources, milestone
2. Typed edges: assigned to, uses, enables, depends on, serves
3. Inspector edits tier, window, skills, permissions, prerequisites
4. Editing bumps the household version and invalidates old plans
5. Backend validation blocks invalid households

Real screenshot of the published prototype, 28 Oct 2026. Amina household, School pickup (T01) selected.

Disruption Injection Studio

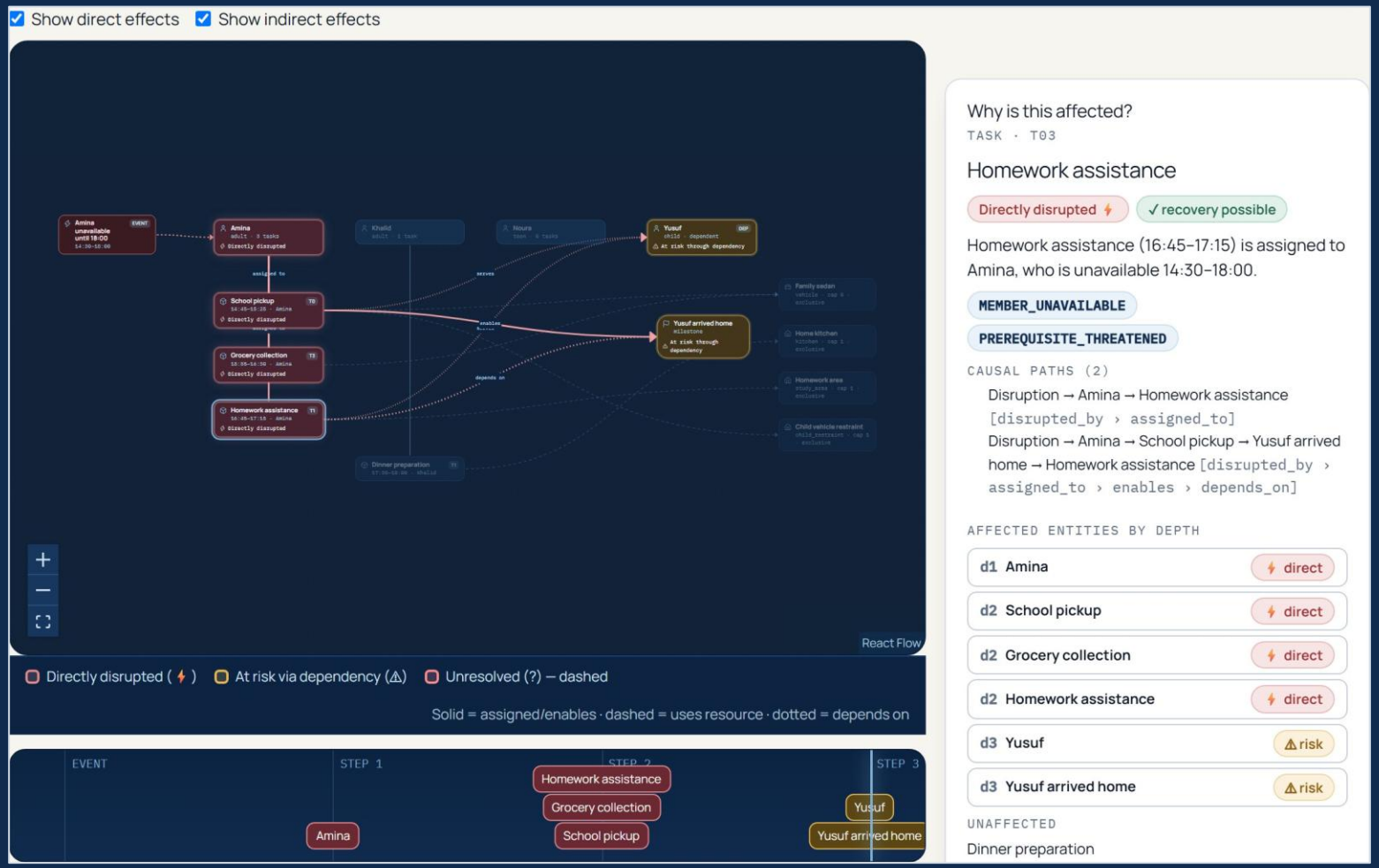


Real screenshots of the published prototype, 28 Oct 2026.

1. Presets are derived from the household itself
2. Manual form works with no AI at all
3. Interpretation → review → confirm: Gemini or the built-in parser
4. Run is disabled until every event is confirmed

Why confirm first? A mistaken interpretation should never silently become a computed plan.

Cascade Laboratory



Real screenshot of the published prototype, 28 Oct 2026. Homework assistance (T03) selected; both causal paths shown.

4
directly disrupted
(Amina + 3 tasks)

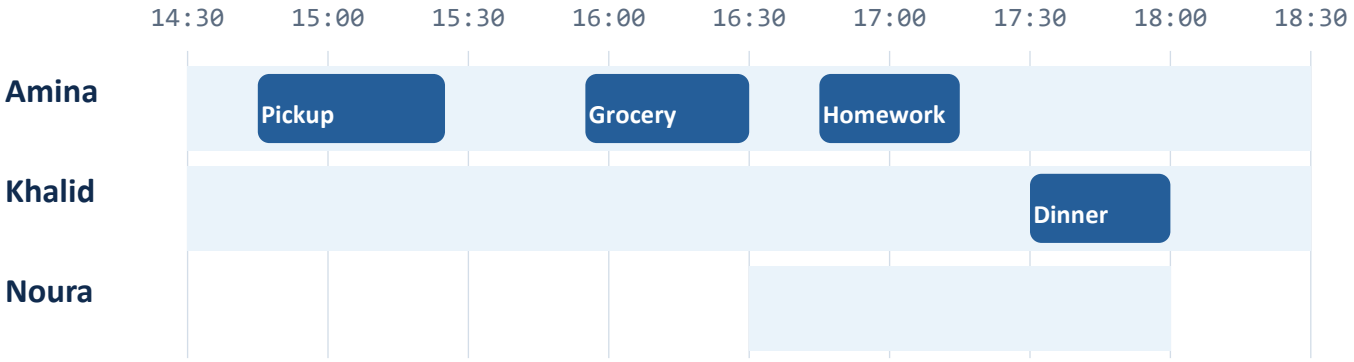
2
at risk through
dependency

3
cascade depth
(scrubbable)

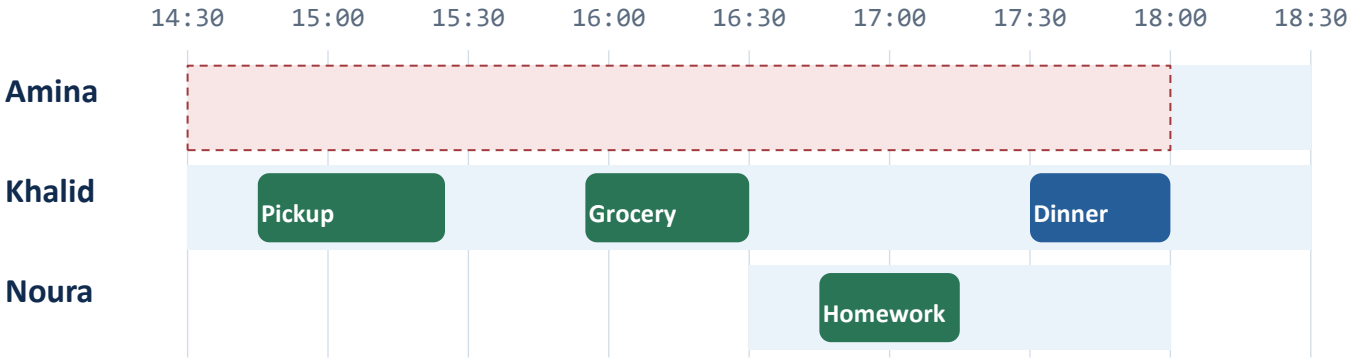
Inspect any node: category, reason codes, every causal path.

Recovery Studio

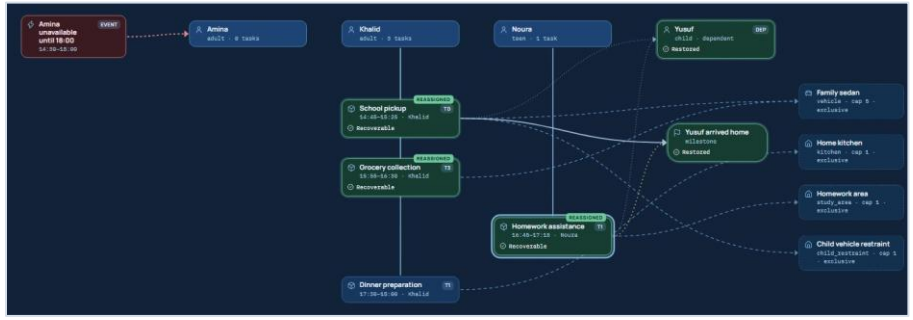
ORIGINAL SCHEDULE



AFTER OPTIMIZATION · OPTIMAL



Green = reassigned and verified

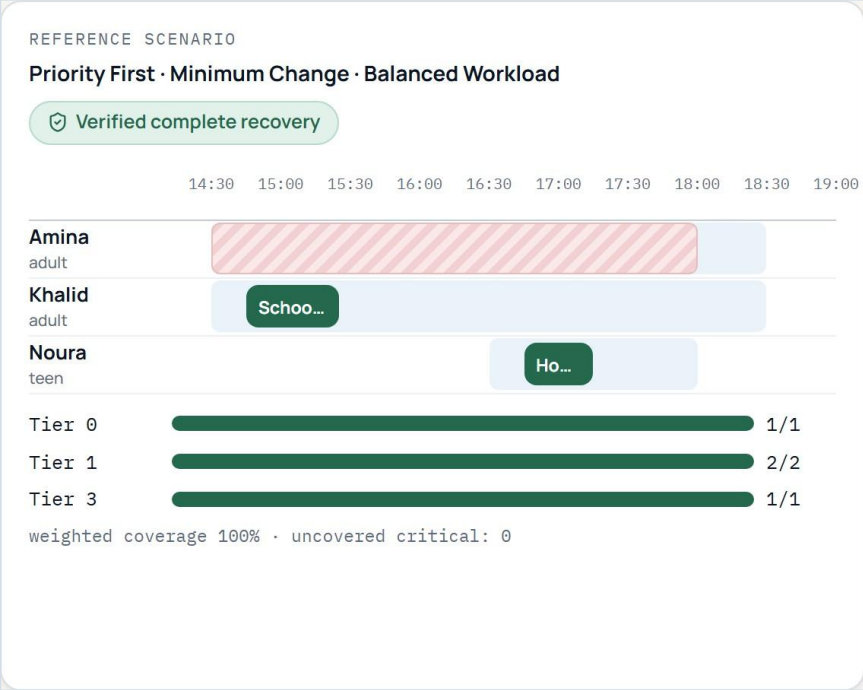


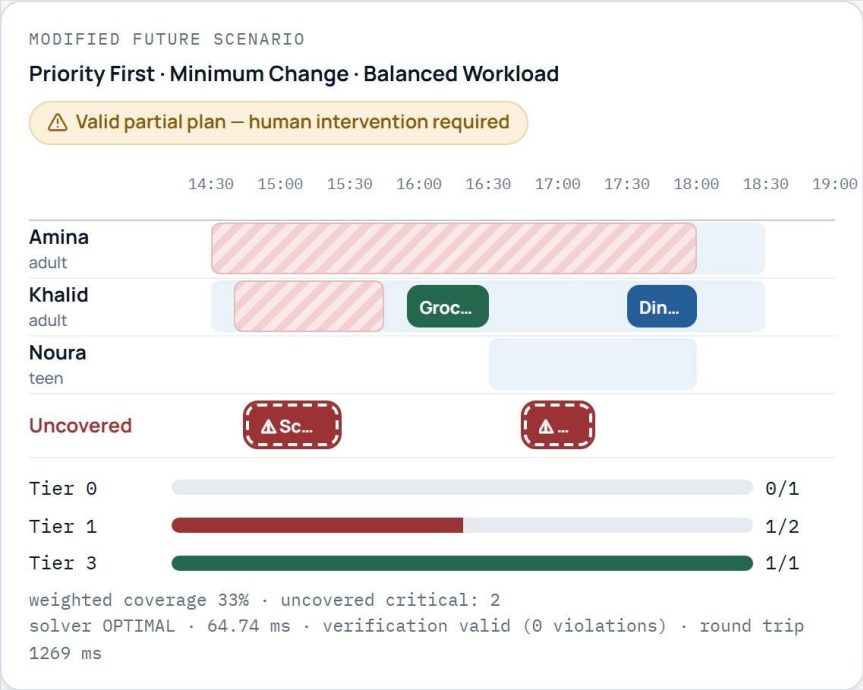
✓ Verified complete recovery

3 reassigned (pickup, groceries → Khalid; homework → Noura) · dinner unchanged · 0 min displacement
Solver OPTIMAL · 51.26 ms (hosted run) · independent check 56/56 passed · 3 strategies agree on 1 arrangement

Source: POST /v1/recovery-plans (live), 28 Oct 2026; graph is a real screenshot, schedules redrawn from the returned assignments.

Family Futures Laboratory





Different result: School pickup, Homework assistance are now uncovered. School pickup cannot be covered: Amina: Amina is not available for any 40-minute slot within 14:40–15:45; Khalid: Khalid is not available for any 40-minute slot within 14:40–15:45; Noura: Noura lacks the skill 'driving'; Noura lacks the permission 'drive'; Noura lacks the permission 'pickup_authorized'; Noura has the restriction 'drive'; Noura has the restriction 'pickup_authorized'. DOMINO will not invent a caregiver, permission or resource.

Change one variable

Caregiver availability
mark a member unavailable

Vehicle availability
take the sedan out

Resource capacity
e.g. seats 5 → 1

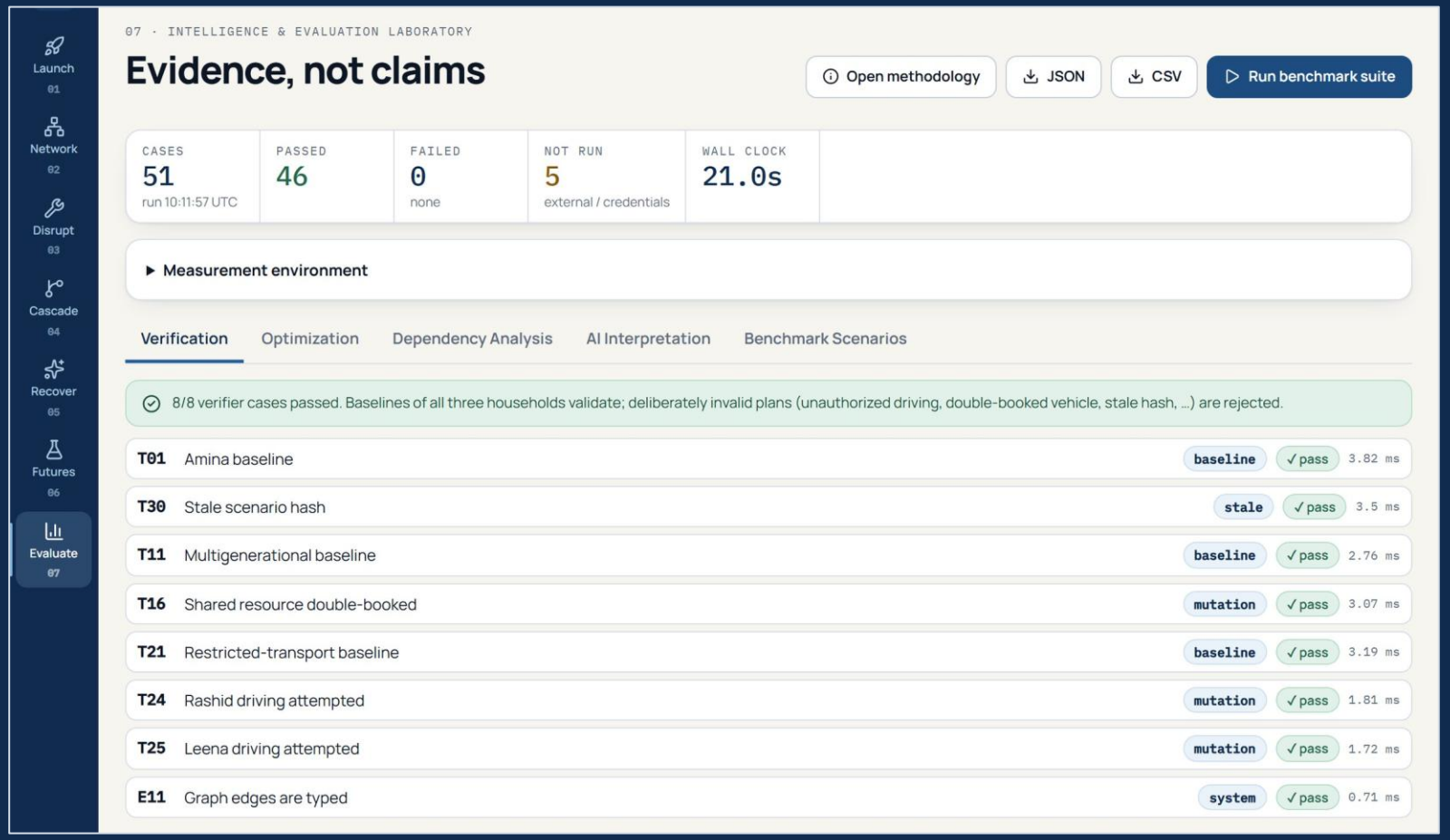
Task priority
change a task's tier

Time-window flexibility
widen or narrow by ± minutes

Every change re-runs graph analysis → CP-SAT → independent verification. Nothing here is a stored alternative.

Real screenshot of the published prototype, 28 Oct 2026: fresh recalculation (solver OPTIMAL, verification 0 violations). Reference preserved on the left.

Intelligence & Evaluation Laboratory



51
cases defined and processed

46
passed

0
failed

5
not run (external / key configured)

Defined ≠ passed: every number comes from a run, never a static page.

Real screenshot of the published prototype after pressing "Run benchmark suite", 28 Oct 2026 (hosted API, Linux, OR-Tools 9.15.6755).

Mathematical optimization with CP-SAT

VARIABLE	x[t,p,s,r] ∈ {0,1} (task, person, start, resources)
COVERAGE	covered[t] = Σ x[t,p,s,r] ≤ 1 (never forced)
PEOPLE	NoOverlap({ [s, s+dur_t) : x[t,p,s,r]=1 }) ∀ p
RESOURCES	exclusive → NoOverlap shared → Cumulative(capacity)
PRECEDENCE	covered[t] ⇒ covered[π] ∧ start[t] ≥ end[π]
OBJECTIVE	lexicographic: max Σ w_t·covered[t] over Tier 0 → fix → Tier 1 → Tier 2 → Tier 3 → then min displacement, changes

Flagship instance (live run)

T01 pickup · Tier 0	Khalid × 6	**14:45**
T02 grocery · Tier 3	Khalid × 24	**15:55**
T03 homework · Tier 1	Noura × 13	**16:45**
T04 dinner · Tier 1	Khalid × 13	**17:30**

56 candidates → chosen start shown →

T01 candidate start times (Khalid)

14:40	**14:45**	14:50	14:55	15:00	15:05

```
passes: Tier0=10 → Tier1=12 → Tier3=2 →
displacement=0 → changes=3
status OPTIMAL · 1 worker · seed 1
```

Source: backend/app/services/{candidate_generation,optimizer}.py; candidate counts and objective values from the live flagship run. Simplified formulation.

Priorities and constraints

HARD CONSTRAINTS — never violated

- ✓ Authorized assignments (skills + explicit permissions)
- ✓ Member availability for the full interval
- ✓ No overlapping responsibilities
- ✓ Resource availability and capacity
- ✓ Dependent safety: only authorized carers
- ✓ Prerequisites complete before dependents

SOFT OBJECTIVES — after coverage

`Tier 0` essential safety & dependent care

`Tier 1` important caregiving

`Tier 2` routine commitments

`Tier 3` deferrable convenience

`then` preserve assignments · least displacement · balance

Lexicographic: optimize Tier 0, fix it, then Tier 1 … so low tiers can never outweigh essentials.

REJECTED CANDIDATE — reason codes

Noura → School pickup ✗

```
MISSING_SKILL
  lacks skill 'driving'
MISSING_PERMISSION
  lacks 'drive'
  lacks 'pickup_authorized'
RESTRICTED
  restriction 'drive'
  restriction 'pickup_authorized'
```

Yusuf: DEPENDENT_MEMBER — never assigned responsibilities.

DOMINO will not invent a caregiver, permission or resource.

Source: candidate_generation.py / explanations.py; reason codes copied from the solver's explanation records (live run).

Independent verification

CP-SAT optimizer
proposes assignments

no shared code

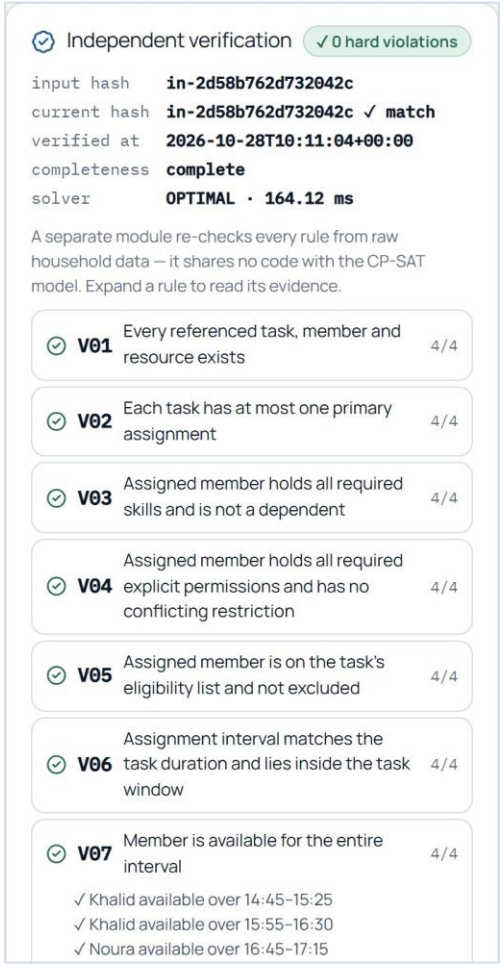
Independent verifier
recomputes every rule from raw data

56 / 56

checks passed · 18 rules · 0 hard violations
(flagship plan, re-run live)

What it checks

- ✓ Eligibility and skills
- ✓ Explicit permissions & restrictions
- ✓ Member availability
- ✓ Scheduling conflicts and travel gaps
- ✓ Resource availability, exclusivity, capacity
- ✓ Dependent-person authorization
- ✓ Prerequisite / dependency completion
- ✓ Scenario version (hash) consistency
- ✓ Human consent remains pending
- ✓ Critical coverage reported accurately



Source: backend/app/services/verifier.py; panel is a real screenshot (28 Oct 2026). Negative tests: backend/tests/unit/test_optimizer_verifier.py, benchmark T16, T24, T25, T30.

The recovery result

Original schedule

4 / 4

responsibilities covered

100% weighted coverage

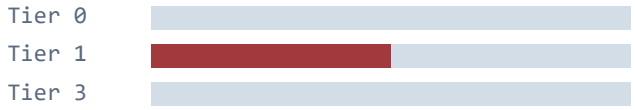


After Amina's disruption

1 / 4

still valid (dinner)

25% weighted coverage



After optimization

4 / 4

covered · 3 reassigned

100% weighted coverage



Coverage ≠ approval

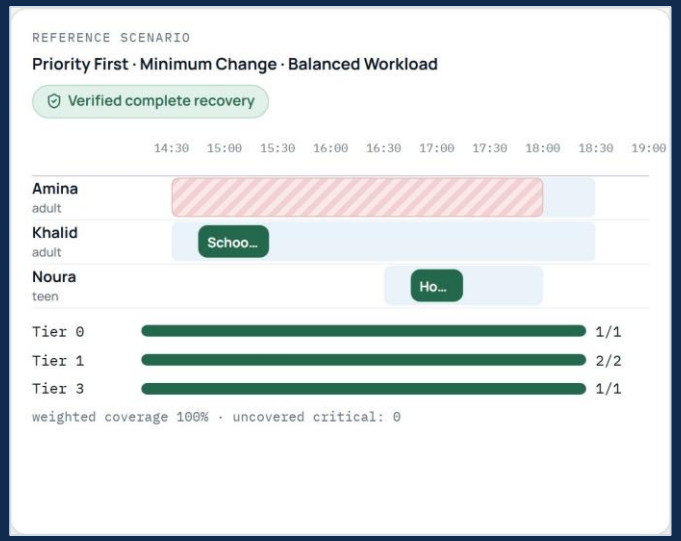
A human must still approve. School pickup needs the new assignee's consent (pending). Approval records an internal simulation receipt only — no one is notified.

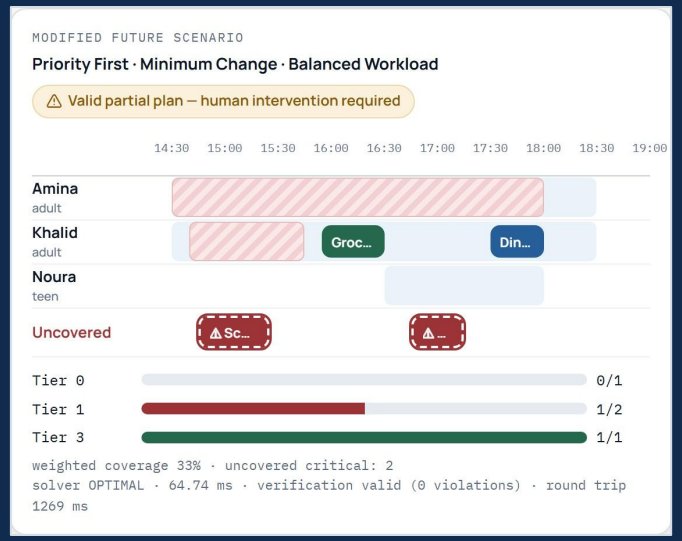
```
Solver, 10 hosted runs: 43–167 ms
median 133 ms · status OPTIMAL · 0 violations
Round trip incl. network:
0.5 s simulate · 0.9 s recover
```

Source: live API runs on 28 Oct 2026 (evidence/live_flagship.json). Free-tier hosting: timings vary; they are measurements, not guarantees.

When recovery is impossible

Different result: School pickup, Homework assistance are now uncovered. School pickup cannot be covered: Amina: Amina is not available for any 40-minute slot within 14:40–15:45; Khalid: Khalid is not available for any 40-minute slot within 14:40–15:45; Noura: Noura lacks the skill 'driving'; Noura lacks the permission 'drive'; Noura lacks the permission 'pickup_authorized'; Noura has the restriction 'drive'; Noura has the restriction 'pickup_authorized'. DOMINO will not invent a caregiver, permission or resource.





Amina out

Verified complete recovery

4/4 covered · 100%
56/56 checks · 0 violations

Amina + Khalid out 14:40–15:45

Valid partial plan — human intervention required

2 critical uncovered: pickup, homework
33% coverage · 30/30 checks
solver OPTIMAL

Why not give the pickup to Noura?

✗ Noura lacks the driving skill

✗ Noura carries restrictions on drive and pickup

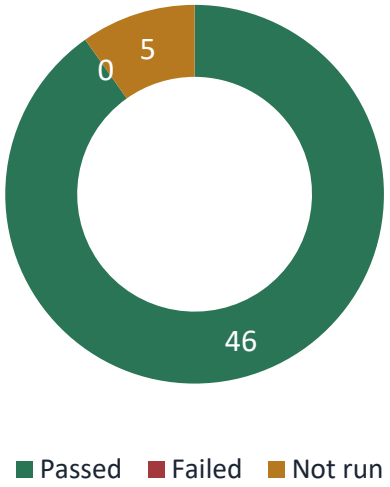
✗ Noura lacks drive and pickup_authorized permissions

✗ Yusuf is a dependent — never assigned

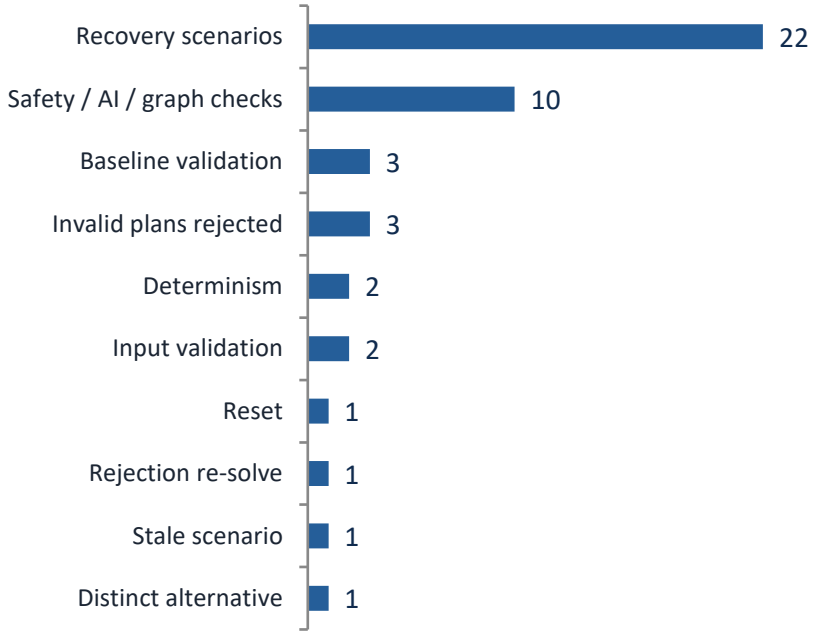
Source: real screenshot + live run 28 Oct 2026. An uncovered critical task is never presented as a complete recovery.

Benchmarking and test evidence

51 cases processed



What the 46 passes cover



Method	Mean weighted coverage	Cases with uncovered critical	Mean changes
DOMINO	71.5%	10 / 21	2.48
Greedy first-fit	71.1%	10 / 21	3.14
No recovery	36.6%	20 / 21	0

DOMINO tied greedy on critical coverage in all 21 cases; it made fewer changes. All baseline outputs also passed the independent verifier.

```
Dependency detection: 11 cases
P 1 · R 1 · F1 1
```

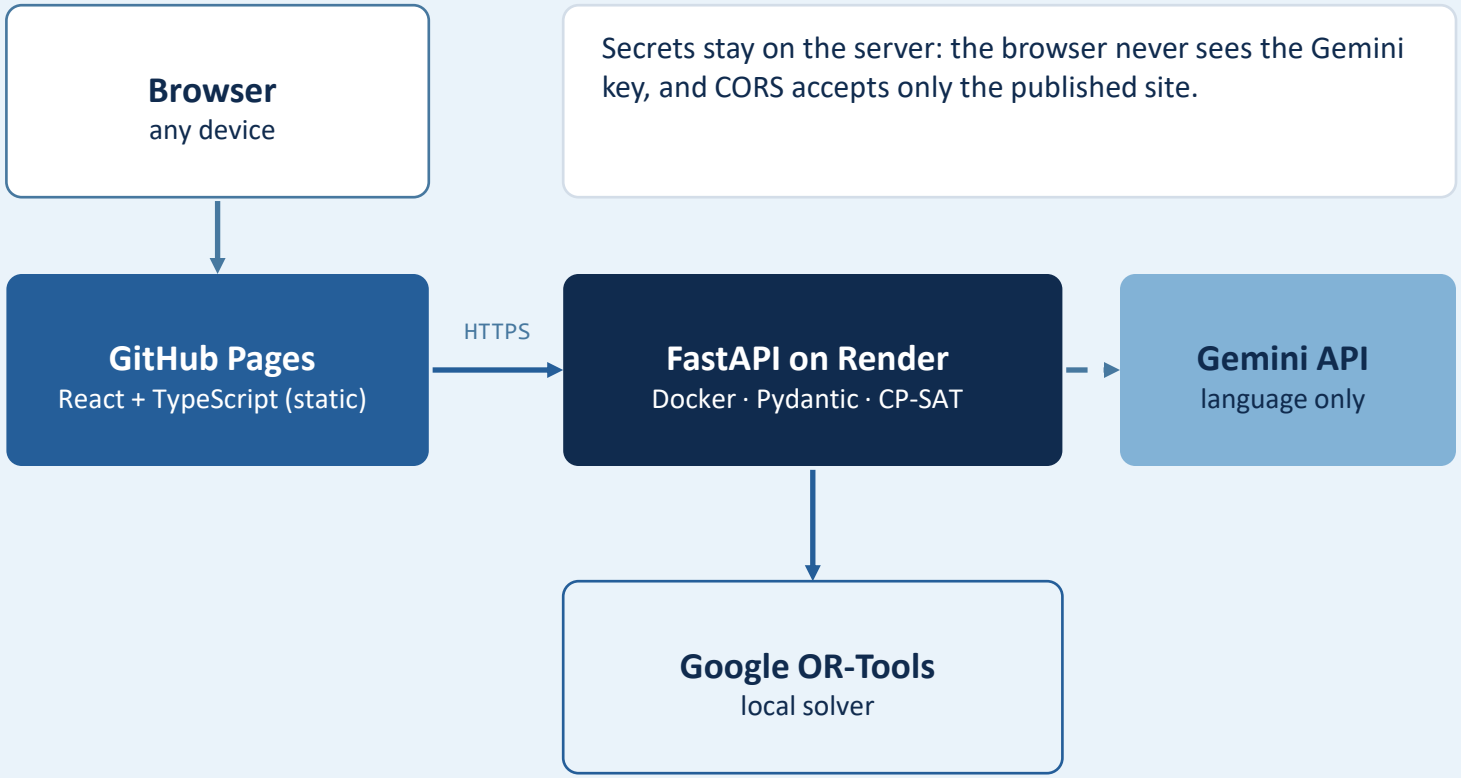
```
Solver per case (n=27): median 187 ms · max 393 ms · case wall: median 801.7 ms, p95 1699.8 ms
```

Categories covered: baseline validation · overlaps · missing permissions · resource unavailability · conflicting schedules · dependent restrictions · secondary disruptions · stale scenario versions.

Source: POST /v1/benchmarks/run on the hosted API, 28 Oct 2026 (Linux, Python 3.12.16, OR-Tools 9.15.6755, 1 worker, seed 1). Synthetic, hand-authored cases; targets are not results.

Security, responsible AI and deployment

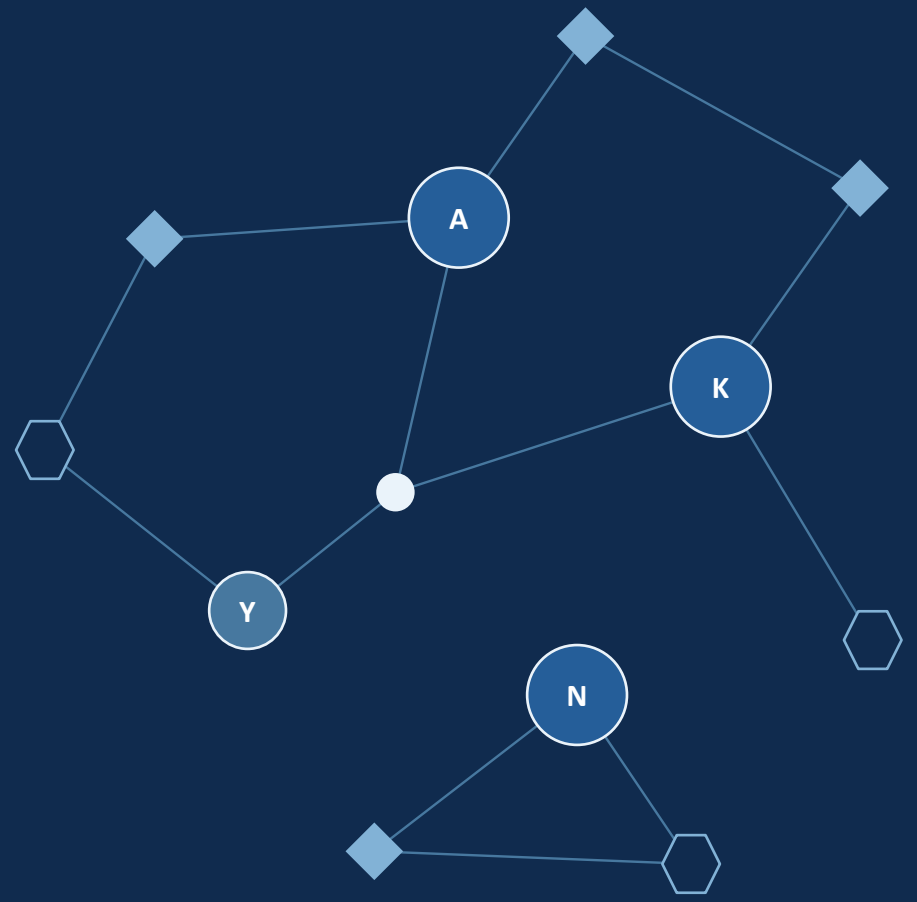
- **Synthetic data only**
  three fictional households
- **Server-side secrets**
  no keys in browser code
- **Input validation**
  Pydantic · size & rate limits · CORS allow-list
- **Permission-aware assignment**
  age never grants permission
- **Human approval**
  simulation receipt; no one is notified
- **Independent verification**
  18 rules, 0 violations required
- **Transparent limitations**
  single day · fictional · no prediction



Status snapshot 28 Oct 2026: /v1/health → ok · OR-Tools CP-SAT · language engines: Gemini + built-in parser. Not an uptime guarantee.

Source: render.yaml, Dockerfile, pages.yml, backend/app/main.py (middleware, CORS, limits). No credentials or private URLs are shown anywhere in this deck.

Family wellbeing and UAE relevance



Conceptual illustration — not data

Potential contribution (not yet measured with real families)

1. More reliable family coordination when plans change
2. Clearer sharing of responsibilities
3. Better understanding of interdependencies
4. Less uncertainty when something goes wrong
5. Transparent decision support, with a human in charge

2026 is the UAE Year of Family. DOMINO relates to the theme only; it is not sponsored or endorsed by any government body. No clinical, psychological or population-level claims are made.

Reference: https://u.ae/en/Year-of-family/Family-year-2026 (UAE Government portal). Benefits are design-level potential, not demonstrated outcomes.

Development status and future potential

IMPLEMENTED AND TESTED	FUTURE POTENTIAL — NOT BUILT
✓ Interactive household modeling and validation	○ More complex households and constraint types
✓ Disruption analysis with causal paths	○ Real travel-time models, multi-day planning
✓ CP-SAT optimization-based recovery	○ Broader accessibility work and audits
✓ Independent 18-rule validation	○ Expanded bilingual (Arabic / English) support
✓ Counterfactual (what-if) recalculation	○ Real-world usability studies with families
✓ Evaluation infrastructure: 51 cases · 46 passed in our run	○ Privacy-preserving data management
✓ Natural language: Gemini + built-in parser, human-confirmed	

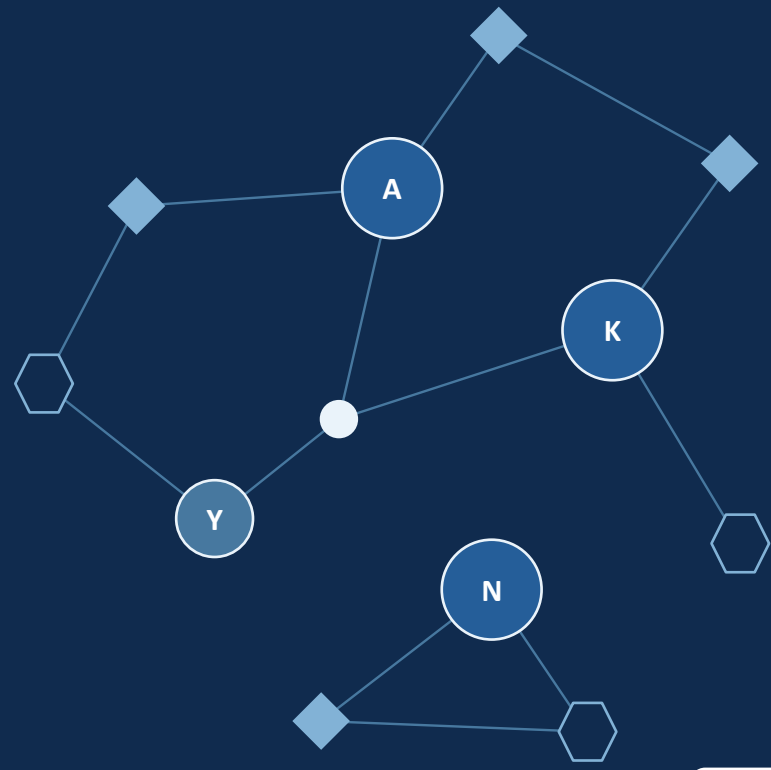
Before real-family use: user studies with consent · privacy & data-protection design · security review · accessibility audits · validation of model assumptions · expert review.
Today: a research-grade prototype on fictional data.

Source: TESTING.md and 28 Oct 2026 benchmark run; future items are plans, not features.

DOMINO

“When one thing changes, everything connects.”

DOMINO helps families understand disruption and calculate a verified way forward.



May Ahmed Mourad · Grade 12
Shamsa Faris Almazrouei · Grade 12
Fatima Mohamed Almazrouei · Grade 11

ADNOC Schools — Madinat Zayed · Abu Dhabi University 6th Annual COE STEM Programming with AI Competition · AI for Family

maymourad.github.io/DOMINO

Scan the QR code or open the link — the live prototype is public